Graphical Abstract

Agri-MetaRL: An Agricultural Meta-Reinforcement Learning Algorithm for Greenhouse Climate Control

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Highlights

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- Greenhouse climate control is modeled as a task distribution over weather, start date, and scenario conditions.
- A new meta-RL algorithm is proposed: MetaAdvantageHead is introduced on Recurrent PPO for task-adaptive advantage correction.
- An evaluation protocol integrating training-distribution and held-out-distribution testing is constructed.
- Agri-MetaRL achieves higher training return (3955) and highest EPI (3.46 $/\mathrm{m}^2$) under the current protocol.
- The method shows competitive temperature control and held-out performance.

Agri-MetaRL: An Agricultural Meta-Reinforcement Learning Algorithm for Greenhouse Climate Control

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Abstract

Greenhouse climate control is an important problem in smart agriculture. The problem has two goals: first, to ensure suitable growing conditions for crops; second, to reduce energy consumption and improve economic returns. Greenhouse systems have three characteristics: strong nonlinearity, time-varying external disturbances, and changing crop growth stages. These characteristics make traditional methods prone to failure. Traditional methods include PID, rule-based control, and MPC. MPC relies heavily on accurate models. Under cross-season and cross-weather conditions, the control performance of these methods tends to degrade. Reinforcement learning (RL) offers a data-driven path for greenhouse control. But most existing studies train a single policy in a single environment. These studies pay insufficient attention to cross-task transfer and fast adaptation.

To address the above issues, this paper formulates greenhouse control as a set of related tasks. These tasks are defined jointly by weather year, start

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date, and scenario conditions. This paper proposes a new meta-reinforcement learning algorithm, Agri-MetaRL. The algorithm does not simply apply off-the-shelf RL to greenhouse control. Instead, it improves the advantage estimation step at the algorithmic level. Based on Recurrent PPO, this paper introduces a support-query style advantage correction mechanism during training before policy updates. Specifically, this paper designs and implements a custom module, MetaAdvantageHead. This module encodes task context from support trajectories and performs task-adaptive correction of generalized advantage estimation on query trajectories during optimization. This paper does not introduce extra policy branches or heavy bi-level optimizers. Under this premise, the learned update mechanism is intended to improve the policy's robustness under task distribution changes.

Experiments were conducted on the GreenLight-Gym greenhouse benchmark environment. Under 5 random seeds, Agri-MetaRL achieved a final mean training return of 3955. This value is higher than Recurrent PPO's 3886 and PPO's 3811. On the held-out weather distribution, for Agri-MetaRL, the mean return per seed for training and test tasks was 3955 ± 16 and 4022 ± 83, respectively. The method maintained competitive held-out performance under the current protocol. Under the fixed protocol, Agri-MetaRL had the lowest violation steps for temperature, humidity, and CO_2. Agri-MetaRL achieved the highest EPI of 3.46 $/m^2$. This value is higher than Recurrent PPO's 3.34 $/m^2$ and PPO's 3.29 $/m^2$.

Keywords: Greenhouse climate control, Reinforcement learning, Meta-reinforcement learning, Recurrent PPO, Smart agriculture

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1. Introduction

Protected agriculture provides stable growing conditions for crops by regulating temperature, humidity, CO_2 concentration, and radiation. This approach can increase yield per unit area, extend supply periods, and resist extreme weather (Van Henten, 2006; Li et al., 2023). In modern high-tech greenhouses, environmental variables, crop state, and actuators are strongly coupled. External weather and crop growth stages change continuously. Thus, greenhouse control has characteristics of strong time-variation, partial observability, and multi-objective optimization. How to achieve high returns, low energy consumption, and strong generalization control while ensuring crop safety bounds is one of the core problems at the intersection of agricultural engineering and intelligent control.

Traditional greenhouse control often uses PID, threshold rules, and manual experience-based strategies (Berenguel et al., 1997; Van Straten et al., 2010). These methods are simple to implement and highly interpretable. But parameters are usually tuned for specific seasons, specific greenhouses, and specific crop stages. When weather distribution changes or crop state shifts, re-tuning is often required. Model predictive control (MPC) can use future predictions and explicit constraints to improve control quality (Camacho and Bordons Alba, 2013). But MPC performance depends on high-precision greenhouse dynamics models and stable solving processes. When model error, parameter uncertainty, and long-horizon rolling optimization coexist, deployment cost remains high. This has been reflected in studies on economic greenhouse MPC and full-cycle crop setpoint optimization (Panagopoulos et al., 2025; Garcia-Manas et al., 2023; van Mourik et al., 2025). RL-guided MPC

and RL-based MPC have been proposed as compromise solutions to balance learning ability and optimization structure (Msaad et al., 2025; Mallick et al., 2025).

In recent years, reinforcement learning (RL) has received wide attention. RL does not rely on precise analytical models and can directly optimize policies toward cumulative returns (Sutton and Barto, 2018; Brockman et al., 2016; Mnih et al., 2015; Arulkumaran et al., 2017; Lillicrap et al., 2015; Haarnoja et al., 2018; Duan et al., 2016a). GreenLight-Gym provides a standardized high-fidelity benchmark environment for greenhouse RL (Van Laatum et al., 2025). Greenhouse RL and deep RL control have been used in recent years for comparing RL with MPC, improving energy efficiency, and introducing interactive control (Morcego et al., 2023; Ajagekar et al., 2023; Xiao et al., 2026). But most existing work limits training to a single or nearly single environment distribution. As a result, the learned policies are often closer to "a strong controller trained on one representative task" than to "a task-aware update structure that transfers across a family of related tasks."

From a learning perspective, changes in weather year, start planting date, parameter perturbations, and scenario coefficients make greenhouse control better viewed as a family of related tasks rather than a single MDP. The goal of meta-reinforcement learning (Meta-RL) is not only to learn a static optimal policy, but also to learn an update structure that transfers across related tasks (Finn et al., 2017; Duan et al., 2016b; Wang et al., 2021; Ravi and Larochelle, 2017). Meta-gradient RL, distributional meta-gradient RL, first-order meta-learning, and comparisons of fine-tuning with Meta-RL all show that learning update rules or adaptation priors can significantly affect

cross-task performance (Xu et al., 2020; Yin et al., 2023; Zhao et al., 2022; Nichol et al., 2018). Inspired by this line of work, this paper proposes a new meta-reinforcement learning algorithm, Agri-MetaRL, and formulates greenhouse control as a Meta-RL problem over weather year, start date, and scenario conditions. The contribution of this paper lies in algorithm design, not merely in applying off-the-shelf RL to greenhouse scenarios. Instead of adding multi-policy branches or constructing heavy inner-outer optimizers, this paper performs task-adaptive correction on advantage estimation. Specifically, task context is extracted from support trajectories, and corrected advantages are generated on query trajectories to guide PPO policy updates. This design aligns with the research thread of improving RL algorithms by learning optimization objectives or update rules (Lu et al., 2022; Oh et al., 2025).

Unlike the traditional "single-environment RL + test-period extrapolation," this paper introduces a task-partitioned training skeleton at the implementation level. Each task segment in the training buffer is split into support and query sets (Vinyals et al., 2016; Snell et al., 2017; Rakelly et al., 2019). The support set encodes context. The query set computes task-adaptive advantages for the auxiliary correction module. This design brings Agri-MetaRL closer to task-conditioned policy optimization while retaining training-flow compatibility with standard PPO and engineering simplicity.

The main contributions of this paper are as follows:

1. For greenhouse climate control, a task-distribution modeling approach over weather year, start date, and scenario conditions is proposed, together with corresponding training and testing protocols for agricul-

tural meta-reinforcement learning.

2. A new meta-RL algorithm design is proposed: task-adaptive correction is introduced at the advantage estimation step, rather than merely applying off-the-shelf RL to greenhouse scenarios.
3. The custom module MetaAdvantageHead is designed and implemented. It performs lightweight task-conditioned advantage correction through support-set context encoding. This module differs from standard Recurrent PPO and is a researcher-developed algorithm component.
4. On the GreenLight-Gym benchmark, a unified evaluation matrix is constructed including learning curves, climate bounds, economic metrics, and held-out evaluation.

2. Materials and Methods

2.1. Simulation Environment

Experiments in this paper were conducted on the GreenLight-Gym greenhouse reinforcement learning platform (Van Laatum et al., 2025; Van Henten, 2006; Farama Foundation, 2024). The platform exposes high-fidelity greenhouse climate dynamics, crop growth, and actuator control through the Gymnasium interface. It supports flexible configuration of weather, parameters, and observation modules. The simulation object is a Venlo-type glass greenhouse common in the Netherlands, with tomato (*Solanum lycopersicum*) as the crop. External weather uses Amsterdam historical weather data. Each episode represents a 60 d growth window. The control step is 15 min. Thus each episode contains 5760 control steps.

2.2. Observation and Action Spaces

The agent's observation vector has 263 dimensions. It consists of six types of information, as shown in Table 1.

Table 1: Composition of the agent observation vector in the GreenLight-Gym tomato greenhouse environment

Component module	Dimension
Indoor climate	4
Crop state	3
Control signals	6
Current weather	5
Time encoding	5
Weather forecast	240

Observations include indoor climate, crop state, current control inputs, current weather, time encoding, and weather forecasts for the next 48 time steps. Running mean and variance were used to normalize observations during training. The action space is a 6-dimensional continuous vector. It corresponds to six actuator types: heating, CO_2 injection, thermal screen, ventilation, supplemental lighting, and shade screen. The action range is $[-1, 1]$. The system updates control inputs in an incremental manner:

$$u_{t+1} = \text{clip}(u_t + a_t \cdot \Delta u_{\max}, u_{\min}, u_{\max}), \tag{1}$$

where $\Delta u_{\max} = 0.1$. The incremental control mechanism can reduce abrupt actuator changes and improve control smoothness and training stability.

2.3. Reward Function and Constraints

This paper adopts the profit-oriented reward function in GreenLightGym. The single-step reward is defined as

$$r_t = \Delta m_{\text{fruit}} p_{\text{fruit}} - (c_{\text{heat}} + c_{\text{CO}_2} + c_{\text{elec}}) - \boldsymbol{\lambda}^\top \boldsymbol{v}_t, \tag{2}$$

where Δm_{fruit} denotes fruit mass increment, $p_{\text{fruit}} = 1.6 \ \text{kg}^{-1}$ is the tomato price, c_{heat}, c_{CO_2}, and c_{elec} denote heating, CO_2, and electricity costs, respectively, and $\boldsymbol{v}_t$ denotes the constraint violation amount. The safe ranges for temperature, relative humidity, and CO_2 are given in Table 2.

Table 2: Greenhouse climate safety bounds (temperature řC, relative humidity %, CO_2 ppm)

Variable	Lower bound	Upper bound
Temperature	15 °C	34 °C
Relative humidity	50 %	85 %
CO_2 concentration	300 ppm	1600 ppm

The penalty weights are $\boldsymbol{\lambda} = [4 \times 10^{-4}, 5 \times 10^{-3}, 7 \times 10^{-4}]^T$. This approach of incorporating safety bounds into policy optimization through penalty terms is consistent with classical practice in constrained reinforcement learning (Achiam et al., 2017; Tessler et al., 2018; Gu et al., 2024).

2.4. Task Distribution and Meta-RL Formulation

We view greenhouse control as a meta-reinforcement learning problem. We define a single control task as $\tau = \{w, s, p, c\}$. In the formula, w is the weather trajectory. s is the start date. p is the greenhouse parameters. c is

the scenario parameters. Weather year and start date cause task differences. So we use 2010 weather as the training distribution $p_{\text{train}}(\tau)$. We randomly select start dates between day 59 and day 96. We use 2011–2013 weather as the test distribution $p_{\text{test}}(\tau)$. We test on days 59, 80, and 100.

For each task segment, we define the support set $\mathcal{D}_\tau^{sup}$ and query set $\mathcal{D}_\tau^{qry}$. The support set encodes task context for the auxiliary correction module. The query set provides the trajectories on which corrected advantages are computed and used for optimization. The objective of Agri-MetaRL is Eq. (3).

$$\max_{\theta,\phi} \ \mathbb{E}_{\tau\sim p_{\text{train}}(\tau)} \left[J_\tau \left(U_\phi(\pi_\theta, \mathcal{D}_\tau^{sup}), \mathcal{D}_\tau^{qry} \right) \right], \tag{3}$$

Here, U_ϕ denotes the support-conditioned advantage correction mechanism used during training, rather than a separate inner-loop policy optimization step. Figure 1 shows the overall system architecture, which consists of three main components and their interactions. First, a task distribution driven by weather year, start date, and scenario conditions defines a family of related control tasks. Second, the greenhouse environment receives control actions and returns observations (state) and rewards at each step. Third, the Agri-MetaRL agent comprises the support set for context encoding, recurrent Actor-Critic networks for policy and value estimation, and the meta-adaptation module for task-adaptive advantage correction. During operation, the agent observes the greenhouse state, selects actions through the policy, and receives rewards from the environment, forming a closed control loop.

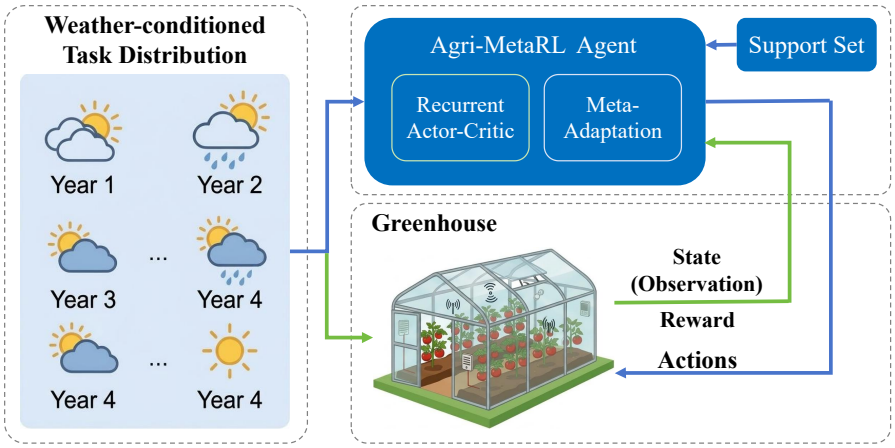

Figure 1: Overall architecture of the greenhouse meta-reinforcement learning control system (Venlo-type greenhouse, tomato, GreenLight-Gym environment), including task distribution, greenhouse environment, Agri-MetaRL agent, and control variables such as temperature, humidity, and CO_2.

2.5. Base Policy: Recurrent PPO

Agri-MetaRL uses Recurrent PPO as the base learner. PPO constrains policy update magnitude through a clipped objective (Schulman et al., 2015a, 2017). GAE provides a low-variance, stable approximation for advantage estimation (Schulman et al., 2015b). Recurrent PPO further introduces LSTM to handle partial observability and temporal dependencies (Ni et al., 2022; SB3-Contrib Contributors, 2024; Hausknecht and Stone, 2015; Pleines et al., 2022). This paper adopts a separated actor-critic LSTM structure. Both the policy and value networks use 1-layer, 256 hidden-unit LSTMs. The policy head fully connected layers are $[1024, 1024]$. The value head is $[128, 128]$. The learning rate is 1.161×10^{-4}, $\gamma = 0.9631$, GAE $\lambda = 0.9666$, batch size is 512, and training epochs are 8. These hyperparameters align with the GreenLight-

Gym benchmark and sb3-contrib Recurrent PPO defaults (Morcego et al., 2023; Van Laatum et al., 2025), with minor adjustments from preliminary grid search to ensure comparability with baseline methods.

2.6. Agri-MetaRL: Support-Query Advantage Correction

Unlike typical Meta-RL methods, MAML (Finn et al., 2017) learns a quickly adaptable initialization through inner-outer optimization loops. PEARL (Rakelly et al., 2019) conditions the policy on latent context variables. Both require extra gradient computation or context inference modules, and policy structure can vary with task. This paper adopts a different design. On the basis of Recurrent PPO, only the advantage estimation step is corrected in a task-aware manner during training. The policy network π_θ remains single and does not switch with task. Meta-learning is embodied in how the advantage signal used for policy gradients is adjusted according to support-derived task context. This design does not require MAML-style inner update loops. It also does not inject context into policy input like PEARL. Instead, support-query partitioning conditions the auxiliary advantage correction module. In this way, the method aims to improve cross-task generalization while maintaining training-flow compatibility with standard PPO and engineering simplicity. The unique contribution of this paper is the introduction of a support-query structure into the advantage estimation step of RL. Through the lightweight MetaAdvantageHead, the method learns how to correct advantages rather than adding task-specific policy branches.

Unlike traditional PPO and Recurrent PPO, which use the same GAE-style advantages for policy updates on all data, Agri-MetaRL splits each task segment into support and query sets. The support set encodes task context

associated with the task variables $\tau = \{w, s, p, c\}$, including weather trajectory, start date, and greenhouse or scenario settings. On the query set, task-relevant correction is applied to the raw advantages. The corrected advantage $\tilde{A}_t = A_t + \delta_\phi(o_t, A_t, c_\tau)$ is then used for policy updates. In this way, the training signal can vary with task context. PPO and Recurrent PPO tend to learn fixed policies on single or nearly single environment distributions. Performance may therefore degrade when task conditions change across weather years, start dates, or related settings. Agri-MetaRL instead models greenhouse control as a task distribution and learns a support-conditioned advantage correction rule during training. This design is intended to improve cross-task robustness without modifying the PPO backbone, adding extra policy branches, or introducing heavy inner-outer optimizers.

Figure 2 illustrates the workflow of the MetaAdvantageHead-based advantage correction mechanism. It contains three modules.

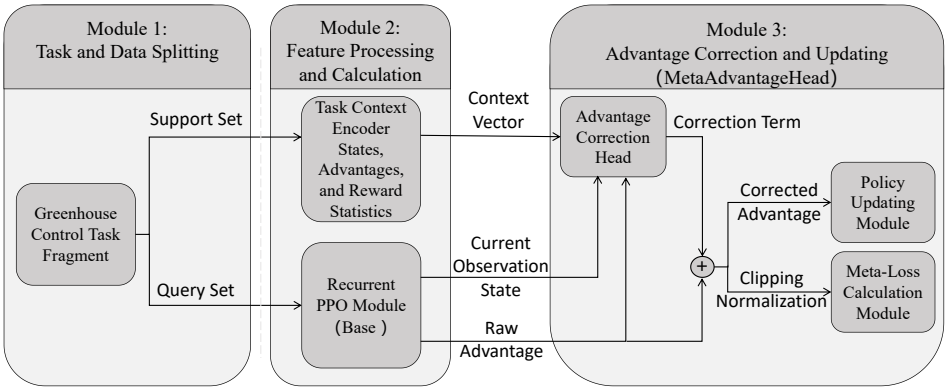

Figure 2: Workflow of MetaAdvantageHead in Agri-MetaRL (based on Recurrent PPO), including task and data partitioning, feature processing and computation, and advantage correction and update modules.

First, each collected task segment is partitioned into support and query parts. The support set ratio is 50% ($meta_support_ratio = 0.5$). The minimum support steps are 256 ($meta_min_support_steps = 256$). Conceptually, this corresponds to an early support portion and a later query portion within the segment. Then, the support set enters the task context encoder. The encoder extracts statistics (e.g., mean and variance) from support set states, advantages, rewards, and returns, and outputs a compact context vector c_τ. Meanwhile, the query set is processed by the base Recurrent PPO module to obtain the current observation o_t and raw advantage A_t. Next, the advantage correction head receives three inputs: context vector c_τ, current observation o_t, and raw advantage A_t. It outputs a correction term through the learnable network δ_ϕ, which is added to the raw advantage:

$$\tilde{A}_t = A_t + \delta_\phi(o_t, A_t, c_\tau), \tag{4}$$

Its implementation contains a context encoder and an advantage correction head. The context encoder takes a 66-dimensional input, consisting of the mean and standard deviation of the first 30 observation dimensions in the support set, together with the mean and standard deviation of support-set advantages, rewards, and returns. It is implemented as two Linear–LayerNorm–SiLU blocks with dimensions $[66 \to 128 \to 64]$, and outputs a 64-dimensional context vector c_τ. For each query step, the current observation o_t is first projected to 64 dimensions by a linear layer. This projected observation, the raw advantage A_t (1 dim), and the context vector c_τ (64 dim) are concatenated into a 129-dimensional input to the correction head. The correction head is a 2-layer MLP, $[129 \to 128 \to 1]$, with SiLU activation in the hidden layer, and outputs a scalar correction term $\delta_\phi(o_t, A_t, c_\tau)$. The

corrected advantage is then normalized and clipped before being used in PPO policy updates. For meta-loss optimization, the target advantage is defined as $G_t - V_\psi(o_t)$ and standardized within the query set, so that the auxiliary module learns task-sensitive corrections toward a return-based target.

$$\mathcal{L}_{meta} = \frac{1}{|\mathcal{D}_\tau^{qry}|} \sum_{t \in \mathcal{D}_\tau^{qry}} (\tilde{A}_t - \hat{A}_t)^2 + \beta \|\tilde{A}_t - A_t\|_2^2, \tag{5}$$

where the target advantage $\hat{A}_t$ is defined as the difference between return and value estimate. It is standardized to zero mean and unit variance within the query set:

$$\hat{A}_t = \frac{(G_t - V_\psi(o_t)) - \mu_{\mathcal{D}_\tau^{qry}}}{\sigma_{\mathcal{D}_\tau^{qry}} + \epsilon}, \tag{6}$$

$G_t = \sum_{k=0}^{T-t} \gamma^k r_{t+k}$ is the discounted return from time t, $V_\psi(o_t)$ is the Critic network's value estimate for the current observation, $\mu_{\mathcal{D}_\tau^{qry}}$ and $\sigma_{\mathcal{D}_\tau^{qry}}$ are the mean and standard deviation of $G_t - V_\psi(o_t)$ within the query set, and ϵ is a numerical stability constant. β is set to 0.05. This design preserves the PPO base structure. By making the corrected advantage approach the return-based target, the auxiliary module is encouraged to learn task-sensitive corrections that are useful for training.

From a mechanistic perspective, support trajectories provide implicit information about the current task, including weather trajectory, start date, greenhouse or scenario settings, diurnal rhythm, and crop stage. The context encoder summarizes these signals from support-set states, advantages, rewards, and returns into a compact task representation c_τ. Conditioned on c_τ, the correction term $\delta_\phi(o_t, A_t, c_\tau)$ varies across tasks, so that the raw advantage on the query set can be adjusted in a task-dependent manner. Consequently, the corrected advantage $\tilde{A}_t$ used in PPO policy gradients re-

flects both the base GAE signal and support-conditioned task information. Through $\mathcal{L}_{meta}$, the auxiliary module is further encouraged to move $\tilde{A}_t$ toward the return-based target $\hat{A}_t$. If the context encoder captures features that are stable across related greenhouse tasks, this mechanism can provide a more task-aware training signal and may contribute to improved robustness on held-out weather distributions, without introducing task-specific policy branches.

2.7. Training and Testing Protocol

Total training steps are 2×10^6. The number of parallel environments is 8. Model selection uses deterministic evaluation. Approximately every 2×10^5 steps, evaluation is run once on the evaluation environment (training distribution: year 2010, day 59). The checkpoint with the highest mean reward is selected for subsequent evaluation. This paper uses a fixed protocol for evaluation. PPO, Recurrent PPO, Agri-MetaRL, and the rule-based baseline are compared on single-episode performance in a deterministic environment. Multi-seed testing is conducted on both training weather distribution and held-out weather distribution.

2.8. Experimental Design and Statistical Analysis

To evaluate method robustness, each algorithm was trained and evaluated with 5 independent random seeds. Seeds were fixed from $\{42, 123, 456, 789, 1024\}$. During training, each episode's start date was randomly sampled within the task definition range. Weather year was partitioned by training/test distribution. During testing, deterministic environment and fixed start days were

used to eliminate evaluation-stage randomness. Performance comparison between methods is reported as mean ± standard deviation over 5 replicates. Given the small sample size, this paper mainly performs descriptive statistics. For inferential tests, paired t-test or Wilcoxon signed-rank test can be used to compare return differences between methods under the same seed.

2.9. Evaluation Metrics

This paper mainly uses the following metrics for evaluation: episode total return, temperature/humidity/CO_2 violation steps, cumulative economic metrics (Revenue, Heat, Electricity, CO_2, EPI), and train-vs-test distribution comparison. To analyze control behavior, this paper also presents indoor climate evolution and key actuator trajectories over the full 60 d cycle.

2.10. Software and Reproducibility

Experiments used Python 3.11, Stable-Baselines3, sb3-contrib, PyTorch, and GreenLight-Gym (Raffin et al., 2021; SB3-Contrib Contributors, 2024; Van Laatum et al., 2025; Brockman et al., 2016; Paszke et al., 2019; Farama Foundation, 2024). Unlike studies that only use off-the-shelf software and hardware, this paper implements a custom MetaAdvantageHead module on top of Recurrent PPO. This module is a researcher-developed algorithm component. It is not included in the standard implementation of Stable-Baselines3 or sb3-contrib. To support paper reproduction, this paper further provides a unified pipeline for figure and table generation, enabling reproducible generation of the figures and data summaries reported in the manuscript.

3. Results

3.1. Learning Curves and Final Returns

Figure 3 shows the mean training learning curves of PPO, Recurrent PPO, and Agri-MetaRL under 5 random seeds. The shaded area indicates one standard deviation across seeds.

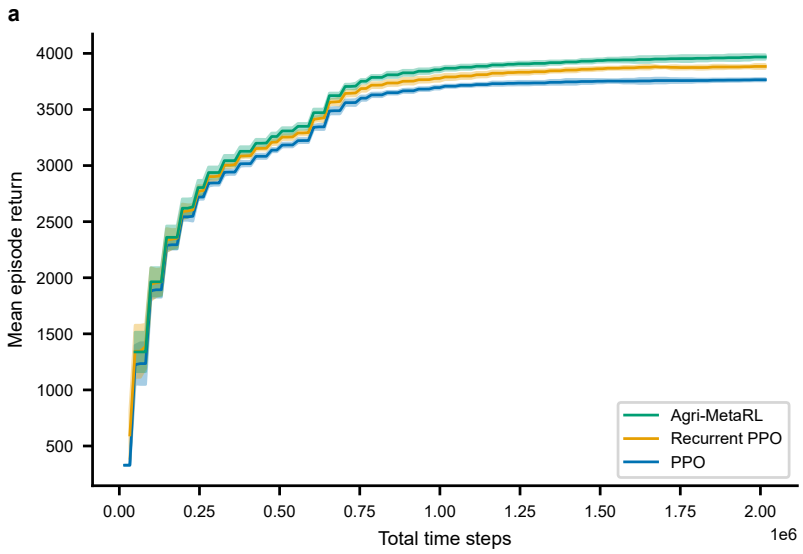

Figure 3: Episode mean return versus training steps for PPO, Recurrent PPO, and Agri-MetaRL under 5 random seeds in the GreenLight-Gym greenhouse environment; shaded area indicates mean ± standard deviation across seeds.

Agri-MetaRL maintains a lead from the early-mid training phase. Recurrent PPO is consistently higher than PPO overall. Table 3 summarizes the final mean returns. Agri-MetaRL reaches 3955 ± 16. This is higher than Recurrent PPO's 3886 ± 32 and PPO's 3811 ± 8. These results suggest that support-query advantage correction improves final performance and is associated with stable behavior across random seeds.

Table 3: Comparison of final mean training returns under 5 random seeds

Method	Final mean episode return
Agri-MetaRL	3955 ± 16
Recurrent PPO	3886 ± 32
PPO	3811 ± 8

3.2. Climate Constraint Performance

Table 4 compares climate violation steps of different methods under the fixed protocol.

Table 4: Comparison of climate violation steps under fixed protocol (mean±SD; unit: steps, 15 min/step)

Method	Temperature	Humidity	CO_2
Agri-MetaRL	97 ± 44	98 ± 29	11 ± 5
Recurrent PPO	110 ± 55	166 ± 33	28 ± 8
PPO	118 ± 28	172 ± 68	33 ± 12

Agri-MetaRL performs best on all three dimensions: temperature, humidity, and CO_2. Violation steps are 97 ± 44, 98 ± 29, and 11 ± 5, respectively. Recurrent PPO is second with 110 ± 55, 166 ± 33, and 28 ± 8. PPO has 118 ± 28, 172 ± 68, and 33 ± 12. Overall, Agri-MetaRL has a clear advantage in climate constraint control.

3.3. Economic Metrics Comparison

Figure 4 shows the cumulative economic metrics results.

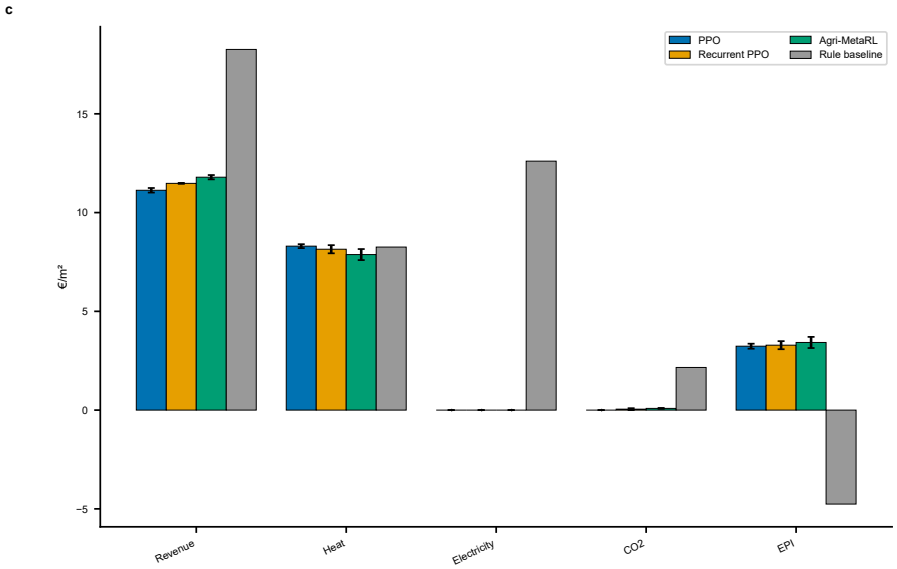

Figure 4: Comparison of cumulative economic metrics (Revenue, Heat, Electricity, CO_2, EPI; unit: $/m^2$) for PPO, Recurrent PPO, Agri-MetaRL, and rule-based baseline under fixed protocol (year 2010, day 59).

Compared with the rule-based baseline, all three RL methods significantly reduce electricity and CO_2 costs and improve overall economic returns. Under multi-seed statistics, Agri-MetaRL has Revenue of 11.56 ± 0.12 $/m^2$, Heat of 8.12 ± 0.32 $/m^2$, CO_2 cost of 0.08 ± 0.02 $/m^2$, and final EPI of 3.46 ± 0.30 $/m^2$. This is higher than Recurrent PPO's 3.34 ± 0.22 $/m^2$ and PPO's 3.29 ± 0.14 $/m^2$. These results suggest that support-query advantage correction can improve the balance between returns and resource consumption. The rule-based baseline has higher Revenue, but its Electricity and CO_2 costs are much larger. Therefore, its overall EPI remains significantly negative (-4.76 $/m^2$).

Table 5 summarizes the comprehensive performance of each controller

on the training and held-out test distributions in the common GreenLightGym benchmark format. Return denotes episode total return. EPI is the Economic Production Index ($/m^2$). TWB_{CO_2}, TWB_T, and TWB_{RH} are the time-within-bounds percentages for CO_2, temperature, and relative humidity, respectively, converted from violation steps over the 5760-step episode. The training distribution used year 2010 weather with fixed start day (day 59). The test distribution used 2011–2013 held-out weather with start days randomly sampled from 59/80/100.

EPI and Return on the test distribution are both higher than on the training distribution. A likely reason is that the test set includes later start days such as day 80 and 100. These runs correspond to the mid-to-late growth season (approximately late March to early June), when day length is longer, heating demand is lower, and photosynthetic potential is higher. Amsterdam weather in 2011–2013 may also provide more favorable radiation and temperature conditions than 2010 in some periods. Therefore, the higher Test EPI should be interpreted mainly as a seasonal evaluation effect rather than a direct indicator of controller superiority.

Table 5: Comprehensive performance comparison of controllers on training and held-out test distributions

	Train (D_{val})					Test (D_{test})				
Controller	Return	EPI	TWB_{CO_2} (%)	TWB_T (%)	TWB_{RH} (%)	Return	EPI	TWB_{CO_2} (%)	TWB_T (%)	TWB_{RH} (%)
Agri-MetaRL	3955 ± 16	3.46 ± 0.30	99.8	98.3	98.3	4022 ± 83	8.34	100.0	98.2	90.1
Recurrent PPO	3886 ± 32	3.34 ± 0.22	99.5	98.1	97.1	3967 ± 77	8.21	100.0	98.0	88.2
PPO	3811 ± 8	3.29 ± 0.14	99.4	98.0	97.0	3921 ± 74	8.13	100.0	97.5	86.3
Rule-based	3094	-4.76	100.0	92.6	36.2	—	—	—	—	—

3.4. 60 d Full-Cycle Control Behavior Analysis

To further observe the long-term regulation characteristics of the controller over the full growth window, Figure 5 shows the actual simulation trajectories of Agri-MetaRL over the full 60 d cycle under the fixed protocol (year 2010, day 59). It includes indoor temperature, relative humidity, CO_2 concentration, and temporal changes in heating, ventilation, and CO_2 injection control signals. The climate subplots show upper and lower safety bounds. This figure was produced from recorded simulation trajectories under the fixed protocol.

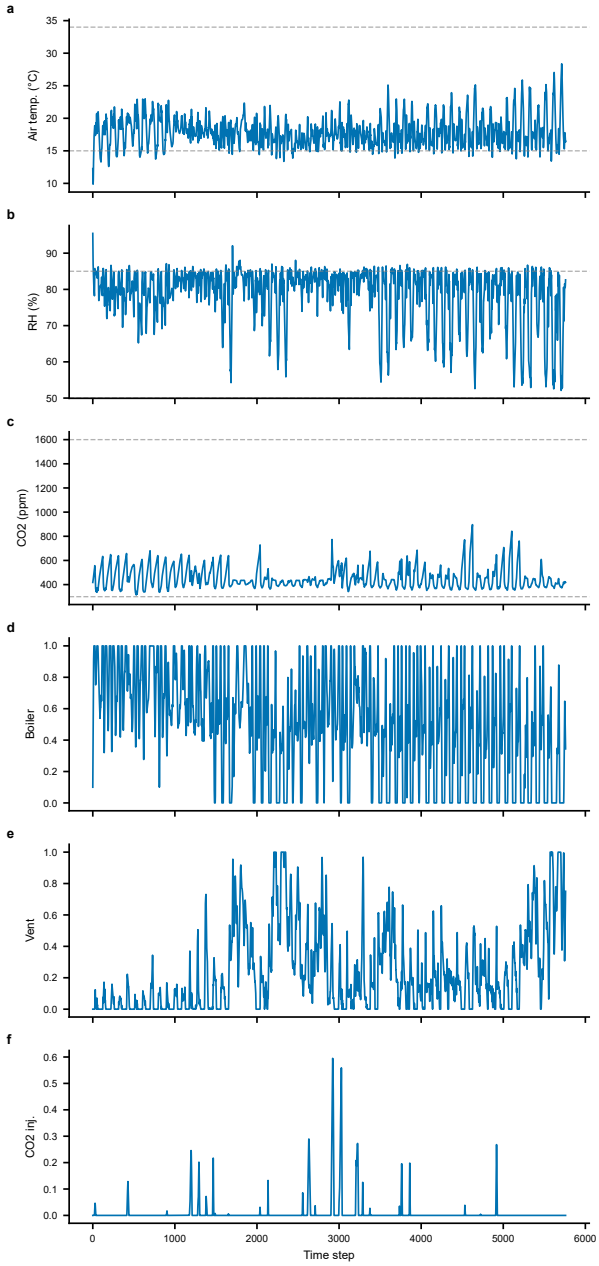

Figure 5: Indoor climate variables and actuator control trajectories of Agri-MetaRL over the 60 d growth window (actual simulation results, fixed protocol year 2010 day 59). Climate variables: temperature (řC), relative humidity (%), CO_2 (ppm). Control inputs: heating, ventilation, and CO_2 injection (dimensionless 0–1). Gray dashed lines indicate upper and lower safety bounds. Horizontal axis is time step (15 min/step).

The trajectories show a clear diurnal pattern over the 60 d horizon. Temperature remains within the safe range for most of the episode, with only brief night-time excursions, which is consistent with the temperature violation statistics reported in Table 4. CO_2 concentration is generally higher during the daytime and decreases during the night, with only a small number of violations on some seeds (11 ± 5 steps in Table 4). The control trajectories indicate coordinated use of heating, ventilation, and CO_2 injection. Heating increases during low-temperature or low-radiation periods, ventilation becomes more active under high-temperature and high-humidity conditions, and CO_2 injection is concentrated mainly during daytime periods when photosynthesis is active. Taken together with the learning curves, climate violation statistics, and economic metrics, these observations suggest that Agri-MetaRL maintains a balanced control behavior over the full 60 d horizon in terms of climate regulation, control smoothness, and return-oriented operation.

4. Discussion

The results of this paper indicate that explicitly modeling greenhouse control as a task distribution and introducing support-query advantage correction on top of Recurrent PPO merits further investigation. Compared with Recurrent PPO, Agri-MetaRL achieves higher final training return under the current protocol. It also attains the highest EPI among the RL baselines and maintains competitive performance on the held-out weather distribution. Because Return and EPI on the held-out distribution are higher than on the training distribution for all three RL methods, this pattern should be

interpreted together with the seasonal differences in the evaluation protocol rather than as direct evidence of superior generalization.

Compared with existing greenhouse RL studies, Morcego et al. (2023) compared DDPG with MPC on the GreenLight model and demonstrated RL potential in greenhouse climate control. Ajagekar et al. (2023) combined deep RL with robust optimization for energy-efficient control of semi-closed greenhouses. Xiao et al. (2026) introduced a human-machine collaborative RL framework with grower participation. These studies mostly focus on single-environment or single-policy optimization. In contrast, this paper explicitly treats greenhouse control as a task distribution over weather year, start date, and scenario conditions, and introduces support-query updates during training. Under the current protocol, the resulting policy maintains competitive performance on held-out weather distributions. GreenLightGym (Van Laatum et al., 2025) provides a standardized benchmark for greenhouse RL, and this paper further constructs a unified protocol including held-out evaluation. This facilitates comparison with subsequent studies.

From the perspective of control objective decomposition, Agri-MetaRL's main advantages are reflected in final training return and economic return (EPI). Relative to Recurrent PPO, Agri-MetaRL achieves higher return, EPI, Revenue, and better climate-constraint performance under the current protocol. In particular, final return increases from 3886 to 3955 (about 1.8%), and EPI increases from 3.34 to 3.46 $/m^2$ (about 3.6%). Agri-MetaRL also records the lowest violation steps on temperature, humidity, and CO_2, which is consistent with the overall ranking in the learning curves and Table 5. These comparisons suggest that the support-query advantage correction mechanism

can improve the balance between return and resource cost while preserving compatibility with the Recurrent PPO backbone and avoiding heavy inner-outer optimization. Relative to PPO, the gains are larger, but Recurrent PPO remains the primary baseline for interpreting the incremental contribution of the proposed method.

At the same time, this paper also observes clear limitations. The results are still mainly based on the current benchmark and task partitioning. Further validation of robustness is needed under richer parameter perturbations, longer time spans, and larger-scale multi-seed experiments. Future work can continue to optimize reward design, context feature construction, and multi-objective trade-offs. From a practical application perspective, Agri-MetaRL's lightweight design (no added policy branches, no heavy inner-outer optimizers) is conducive to integration into existing greenhouse control systems. The trade-off between support set length and computational cost still needs attention. In addition, simulation-to-reality transfer requires further validation in real greenhouses, especially for sensor noise, model bias, and delay.

This paper should also be viewed as an engineering prototype for Meta-RL-oriented greenhouse control, not as the final answer to the greenhouse meta-learning problem. The current implementation already has a clear support/query meta-update skeleton and a reproducible experimental and visualization workflow. But to form stronger statistical evidence, extension experiments are still needed with more seeds, richer task distributions, and more complete multi-baseline comparisons.

5. Conclusion

This paper proposes Agri-MetaRL, an agricultural meta-reinforcement learning algorithm for greenhouse climate control. The method formulates greenhouse control tasks as a task distribution over weather year, start date, and scenario conditions. On the basis of Recurrent PPO, it introduces the support-query MetaAdvantageHead. Support-derived task context is used to correct query advantages during training. Compared with traditional single-environment RL, Agri-MetaRL places more emphasis on a cross-task transferable update structure.

Under the current protocol, Agri-MetaRL achieved higher final training returns. It also maintained relatively stable performance on the held-out weather distribution. Climate constraint analysis and economic metrics comparison further support the effectiveness of task-context encoding and advantage correction. Overall, the results indicate that Agri-MetaRL provides a practical and extensible framework for Meta-RL-oriented greenhouse climate control research.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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